

Utkarsh Yadav

+91-7007663578 | utkarshxp19@gmail.com | karsh.dev | github.com/Rytnix

EDUCATION

IIMT Engineering College

Bachelor of Technology in Computer Science Engineering, GPA: 8.57/10.0

Meerut, India

July 2019 – June 2023

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL, Bash
Frameworks & Libraries: OpenCV, Scikit-learn, NLTK, SpaCy, TensorFlow, Keras, Flask, Springboot
Tools & Technologies: Kubernetes, Docker, Git, PostgreSQL, MySQL, SQLite, LAMP Stack, N8N
Cloud Platforms: AWS, Google Cloud Platform, IBM Cloud
Operating Systems: Linux, Raspberry Pi, NVIDIA Jetson Nano

PROFESSIONAL EXPERIENCE

Proeffico Solutions Private Limited

Software Development Engineer - ML & Gen AI

Noida, India

March 2024 – Present

- Facial Recognition System Optimization:** Redesigned live facial recognition pipeline processing RTSP video streams, reducing latency from 40+ seconds to under 1 second through architectural improvements and parallel processing implementation
- High-Performance Embedding Retrieval:** Engineered Redis-based vector storage system retrieving facial embeddings from 100K+ database in under 200ms, implementing efficient indexing and caching strategies for real-time identification
- Anti-Spoofing Integration:** Developed and integrated liveness detection module for video and photo inputs, preventing spoofing attacks and enhancing system security for client authentication workflows
- Computer Vision API Development:** Built production-grade REST APIs for facial recognition, demographic analysis, and biometric verification, handling 1000+ daily requests with 99.5% uptime
- Gen AI Document Generation System:** Architected self-hosted solution blueprint generator for Oman-based client, fine-tuning model on RSD documentation to achieve 70% accuracy in generating technical specification documents
- Team Leadership & Innovation:** Led cross-functional team of 5 engineers, proposing and implementing Gen AI-powered fire detection system and sentiment analysis automation
- Real-Time Video Intelligence:** Currently developing CCTV video frame analysis agent enabling natural language querying of live surveillance feeds for security and monitoring applications

Proeffico Solutions Private Limited

Software Engineer Intern

Noida, India

September 2023 – March 2024

- Mutual Fund Platform Backend:** Designed and implemented backend infrastructure for Edhanam mutual fund platform serving Striment Technologies, integrating StarMF API for real-time fund data and transaction processing
- API Performance Optimization:** Reduced mutual fund API response times by 80% (from 3-4 seconds to under 800ms) through database query optimization, Redis caching implementation, and connection pooling strategies
- Database Architecture:** Implemented multi-layer caching mechanism and database indexing strategies, significantly improving query performance for high-frequency fund data retrieval operations
- Third-Party Integration:** Successfully integrated StarMF APIs handling fund discovery, NAV updates, transaction processing, and portfolio management for seamless client experience

PROJECTS & TECHNICAL IMPLEMENTATIONS

Symptom-to-Disease Prediction System

Gen AI

Personal Project

2025

- Hybrid AI Architecture:** Designed two-stage pipeline integrating traditional ML classification with large language model reasoning layer to analyze symptoms and generate disease predictions with explanatory context
- Model Optimization:** Fine-tuned 3B parameter open-source language model for medical domain, creating custom training dataset through data augmentation and medical literature extraction
- Performance Enhancement:** Achieved 50% accuracy improvement (60% baseline to 90% post-training) through systematic dataset curation, hyperparameter tuning, and transfer learning techniques
- Clinical Recommendations:** Implemented contextual reasoning system generating personalized precautions and treatment suggestions based on predicted conditions and patient symptom profiles

AI-Powered Trend Intelligence Platform

Agentic AI

Personal Project

2025

- **Data Collection Pipeline:** Architected automated web scraping system extracting real-time content from social media platforms and news sources, processing 10,000+ daily data points
- **NLP Classification Engine:** Implemented transformer-based topic categorization using pre-trained embeddings to classify content across multiple domains with 85% accuracy
- **Predictive Analytics:** Developed time-series forecasting model analyzing engagement metrics and growth patterns to identify trending topics 24-48 hours before mainstream adoption
- **Interactive Visualization:** Created dynamic dashboard displaying trend trajectories, category distributions, and prediction confidence scores

Self-Hosted Research Paper Query System

Portfolio Enhancement

NLP

2024

- **Document Intelligence Agent:** Built conversational AI system enabling natural language querying across research paper collection, eliminating need for manual document review
- **Vector Database Implementation:** Engineered custom embedding pipeline transforming research papers into vector representations stored in local vector database, ensuring complete data privacy
- **Semantic Search Engine:** Implemented retrieval-augmented generation (RAG) architecture providing accurate answers with section-level citations and contextual excerpts
- **Self-Hosted Infrastructure:** Deployed entirely on local infrastructure without external API dependencies, utilizing open-source LLM and embedding models for complete autonomy
- **User Experience:** Integrated chatbot interface into portfolio website, reducing research paper navigation time by 80% and improving visitor engagement

PUBLICATIONS & TECHNICAL DOCUMENTATION

Scalable Facial Recognition Architecture for Large-Scale Surveillance Systems: Authored comprehensive technical documentation on designing and implementing distributed facial recognition infrastructure supporting 1,000+ concurrent RTSP camera streams. Detailed multi-threading and multi-processing strategies, load balancing techniques, and real-time processing optimization. Published in Proeffico Solutions internal engineering documentation (2025)

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Open Source Contributions (2023):** Contributed optimized Travelling Salesman Problem algorithm to open-source repository using dynamic programming techniques
- LeetCode Profile:** Rating 1453 with 150+ algorithmic problems solved across data structures, algorithms, and system design
- MesoHacks Hackathon (2022):** Secured Top 10 position among 2,500+ participants in national-level hackathon
- Get-Set Hack by RISE (2022):** Ranked 50th among 22,000+ participants in competitive coding hackathon
- CodeChef October Long Challenge (2022):** Achieved Global Rank 231 out of 10,000+ participants (Top 2%)
- CodeChef August Long Challenge (2022):** Achieved Global Rank 591 out of 10,000+ participants (Top 3%)

LEADERSHIP & STRATEGIC INITIATIVES

ML & Gen AI Team Lead

Proeffico Solutions Private Limited

Leading Cross-Functional Engineering Team

March 2024 – Present

- **Team Management:** Lead team of 5 engineers across machine learning, computer vision, and generative AI projects, coordinating sprint planning, code reviews, and technical architecture decisions
- **Client-Focused Innovation - CSI Analytics:** Proposed region-based Customer Staff Interaction (CSI) monitoring system leveraging computer vision to capture actionable insights from customer-staff engagements across retail locations, enabling data-driven service improvement strategies
- **Security Guard Monitoring Solution:** Designed automated guard activity detection system identifying sleeping or inattentive behavior through pose estimation and behavioral pattern analysis, reducing security incidents and improving workplace safety compliance
- **Gen AI Fire Detection Pipeline:** Architected two-stage fire detection confirmation system combining traditional computer vision with generative AI verification layer, reducing false positives by 85% while maintaining real-time response capabilities
- **Strategic Product Development:** Identified market gaps and proposed AI-driven solutions resulting in three new product offerings, directly contributing to client acquisition and revenue growth